Alben Rome B. Bagabaldo

Research Data Scientist | Applied ML | Real-World Data Analysis
New Brunswick / Piscataway, NJ | Open to New York / hybrid roles

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PROFESSIONAL SUMMARY

Ph.D.-trained research data scientist with experience developing hypotheses, designing rigorous analyses, building machine learning and statistical models, and extracting meaningful signals from complex real-world datasets. Strong background in applied prediction, simulation, geospatial data, transportation systems, literature synthesis, uncertainty analysis, and stakeholder-facing research translation. Experienced in turning open-ended technical questions into reproducible Python and SQL workflows, measurable metrics, interpretable models, and clear decision-ready findings for engineering, research, and public sector collaborators.

CORE SKILLS

- **Research Design:** hypothesis development, literature review, methodology synthesis, experimental design, theory testing
- **Applied Statistics:** exploratory data analysis, uncertainty analysis, sensitivity analysis, hypothesis testing, error analysis, model diagnostics
- **Machine Learning:** supervised learning, regression, random forests, gradient boosting, clustering, feature engineering, model validation
- **Prediction Problems:** traffic flow prediction, anomaly-aware prediction, demand and service impact analysis
- **Programming/Data:** Python, SQL, R; pandas, NumPy, SciPy, scikit-learn, GeoPandas, NetworkX, OSMnx, Jupyter, Git
- **Simulation:** Monte Carlo simulation, scenario generation, traffic simulation, what-if analysis, synthetic/real data integration
- **Real-World Data:** geospatial data, traffic sensors, GTFS, road networks, location-based services data, raster/vector workflows
- **Communication:** technical writing, stakeholder briefings, cross-functional collaboration, mentoring, publication quality synthesis

EXPERIENCE

2025– **Rutgers University – Center for Advanced Infrastructure & Transportation** (Piscataway, NJ)
Postdoctoral Associate

- Developed research questions, metrics, and reproducible Python workflows for evaluating battery-electric bus resilience under disrupted transportation-network conditions.
- Integrated General Transit Feed Specification data, road networks, flood scenarios, garage assignments, route structures, energy assumptions, and operational constraints into analysis-ready datasets.
- Conducted scenario analysis and sensitivity testing to compare competing assumptions and translate complex operational results into decision-support outputs for infrastructure and transit stakeholders.

2024–25 **Princeton University – Complex Infrastructure Systems Group** (Princeton, NJ)
Postdoctoral Research Associate

- Led applied data science research on digital twins for intelligent intersections, focusing on real-world mobility data, traffic sensing, simulation, AI/ML methods, and safety-oriented analytics.
- Developed probabilistic and Monte Carlo simulation workflows to test hypotheses about pedestrian-vehicle interaction risk under different traffic, signal, and behavioral conditions.
- Translated ambiguous safety and mobility questions into measurable research designs, validation logic, technical reports, and stakeholder-facing outputs for university and local-government collaborators.

2019–24 **University of California, Berkeley** (Berkeley, CA)
Ph.D. Researcher, Mobile Sensing Lab / HumNet Lab; Graduate Student Instructor

- Designed large-scale microscopic traffic simulation experiments to test hypotheses about how navigation information and routing behavior affect congestion formation, propagation, dissipation, and network-level performance.
- Built reproducible analytics pipelines using structured experiment definitions, documented parameters, traffic flow outputs, and consistent evaluation metrics across simulation scenarios.
- Developed machine learning models to predict traffic flow under faulty detector conditions, covering data cleaning, missing-data handling, feature construction, clustering, model selection, validation, and interpretable error analysis.
- Conducted interdisciplinary research with civil engineering, electrical engineering, computer science, and data science collaborators on mobility systems, traffic sensing, simulation quality, and real-world transportation data.
- Served as Graduate Student Instructor for quantitative and data-oriented coursework, strengthening technical communication, mentoring, and explanation of complex concepts to diverse learners.

2014–19 **Mapúa University & Mapúa Phil-LiDAR Program** (Manila, Philippines)
Research Associate / GIS Specialist; Senior Science Research Specialist; Instructor & Program Chair duties

- Built repeatable geospatial data-processing workflows for large LiDAR-derived terrain, flood-hazard, infrastructure, and planning datasets.
- Produced standardized raster and vector data products with documented QA/QC, projections, tiling, metadata, and stakeholder-ready deliverables.
- Coordinated technical handoffs, trainings, and documentation for government and infrastructure users of complex spatial data products.
- Taught civil engineering and transportation topics while supporting program-level curriculum, assessment, accreditation, and industry engagement.

SELECTED RESEARCH DATA SCIENCE PROJECTS

Hypothesis-Driven Study of Navigation Apps and Congestion Dynamics

- Developed and tested hypotheses about how static and dynamic routing behavior affects congestion formation and spread across transportation networks.
- Used microscopic traffic simulation, scenario design, and network-level metrics to evaluate travel time, speed, density, queue formation, delay, and congestion dissipation.
- Relevant strengths: independent research, real-world behavioral motivation, rigorous inquiry, simulation-based testing, reproducible workflows, and clear communication of complex findings.

Probabilistic Safety Simulation for Pedestrian-Vehicle Interactions

- Developed probabilistic and Monte Carlo simulation workflows to evaluate pedestrian-vehicle risk under different crossing, signal, proximity, and traffic conditions.
- Converted ambiguous safety concepts into measurable variables, scenario definitions, uncertainty-aware outputs, and stakeholder-facing interpretation.
- Relevant strengths: hypothesis testing, simulation, uncertainty quantification, safety analytics, and research translation.

Network Disruption and Rerouting Analysis for Transit Resilience

- Integrated GTFS, road networks, hazard layers, route assignments, and operational assumptions to evaluate how disruptions alter service, accessibility, rerouting burden, and energy demand.

- Developed multi-criteria metrics to compare direct exposure, accessibility loss, ridership impact, garage criticality, and operational energy effects across scenarios.
- Relevant strengths: heterogeneous real-world datasets, spatial analytics, scenario analysis, metric design, and communication of decision-ready findings.

EDUCATION

Ph.D. **University of California, Berkeley**
Civil and Environmental Engineering, conferred August 2024
Dissertation: "Navigating Urban Traffic: From Data to Simulations to Real-World Impacts"
Adviser: Prof. Alexandre M. Bayen *Committee:* Prof. Marta C. Gonzalez, Prof. Joan Walker
Relevant coursework: Scalable Spatial Analytics; Behavior Modeling; Applications in Data Analysis; Transportation Data Analysis; Sensing and Signal Interpretation; Energy Systems and Control; Public Transportation Systems

M.S. **Mapúa University**
Civil Engineering, Transportation Engineering, August 2016

B.S. **Southern Luzon State University**
Civil Engineering, *cum laude*, March 2012

SELECTED EVIDENCE OF EXCELLENCE

- First-author paper, "Impact of Navigation Apps on Congestion and Spread Dynamics on a Transportation Network", published in *Data Science for Transportation*; uses microscopic traffic simulation to evaluate static and dynamic routing behavior.
- Co-author, "The Lord of the Ring Road: A Review and Evaluation of Autonomous Control Policies for Traffic in a Ring Road", published in *ACM Transactions on Cyber-Physical Systems*; cited 30 times.
- UC Berkeley Outstanding Graduate Student Instructor Award, 2024.
- Google Cloud Platform Credits, USD 10,000, for AI and digital twin experiments, 2025.

SELECTED PUBLICATIONS AND PREPRINTS

- **Bagabaldo, A. R.**, Gan, Q., Bayen, A. M., & Gonzalez, M. C. (2024). Impact of navigation apps on congestion and spread dynamics on a transportation network. *Data Science for Transportation*, *6*, Article 12. doi:10.1007/s42421-024-00099-w
- Chou, F.-C., **Bagabaldo, A. R.**, & Bayen, A. M. (2022). The lord of the ring road: A review and evaluation of autonomous control policies for traffic in a ring road. *ACM Transactions on Cyber-Physical Systems*, *6*(1), 1–25. doi:10.1145/3494577
- Tang, Y., Alhadlaq, A., **Bagabaldo, A. R.**, & Gonzalez, M. C. (2025). Designing transit routes based on vehicle routing behavior determined through location-based services data. *EPJ Data Science*, *14*, Article 45. doi:10.1140/epjds/s13688-025-00559-5
- **Bagabaldo, A. R.**, & Hackl, J. (2025). Digital twins for intelligent intersections: A literature review. *arXiv preprint*. arXiv:2510.05374
- **Bagabaldo, A. R.**, & Hackl, J. (2025). Improving pedestrian safety at intersections using probabilistic models and Monte Carlo simulations. *arXiv preprint*. arXiv:2503.07805

Updated May 2026